



AtliQ Motors – EV Market Analysis

India Electric Vehicle Market Entry Study | FY2022–FY2024

Project type: Data Analytics / Business Intelligence portfolio project

Tools: Python, Pandas, MySQL, Power BI, DAX, Power Query

Objective: Analyze India's EV market, manufacturer competition, state-level adoption, growth and future opportunity to support AtliQ Motors' market-entry decision.

1. Executive Summary

Total EV sales increased from 271,150 units in FY2022 to 1,019,593 units in FY2024, an overall increase of 276.0%. Two-wheelers represented the majority of EV sales, while four-wheeler sales also expanded substantially.

The analysis combines manufacturer-level and state-level sales data to study competition, penetration, growth, seasonality and state opportunity.

2. Business Questions

1. Top and bottom 2-wheeler manufacturers for FY2023 and FY2024.
2. Top 5 states by EV penetration for 2-wheelers and 4-wheelers in FY2024.
3. States with negative EV penetration change from FY2022 to FY2024.
4. Quarterly 4-wheeler sales trends for the top 5 manufacturers.
5. Delhi vs Karnataka EV sales and penetration in FY2024.
6. CAGR of 4-wheeler units for the leading manufacturers.
7. Top 10 states by EV sales CAGR from FY2022 to FY2024.
8. Peak and low EV sales months.
9. 2030 EV sales projection for the top states by FY2024 penetration.
10. Revenue growth analysis for 2W and 4W using average unit-price assumptions.

3. Data & Methodology

The project uses a cleaned date dimension, manufacturer sales table and state sales table. The state table is linked to the date dimension by date to derive fiscal year and quarter.

Data preparation: missing-value checks, date standardization, text cleaning, data-type validation and preparation for SQL/Power BI analysis.

EV penetration: $\text{EV sales} \div \text{total vehicle sales} \times 100$.

CAGR: $(\text{FY2024 sales} \div \text{FY2022 sales})^{(1/2)} - 1$.

4. Manufacturer Analysis

FY2023 and FY2024 2-Wheeler Leaders

FY	Group	Rank	Manufacturer	Sales
FY2023	Top 3	1	OLA ELECTRIC	152,583
FY2023	Top 3	2	OKINAWA	96,945
FY2023	Top 3	3	HERO ELECTRIC	88,993
FY2023	Bottom 3	1	JITENDRA	8,563
FY2023	Bottom 3	2	BEING	11,018
FY2023	Bottom 3	3	PURE EV	11,556
FY2024	Top 3	1	OLA ELECTRIC	322,489
FY2024	Top 3	2	TVS	180,743
FY2024	Top 3	3	ATHER	107,552
FY2024	Bottom 3	1	BATTRE ELECTRIC	4,841
FY2024	Bottom 3	2	REVOLT	7,254
FY2024	Bottom 3	3	KINETIC GREEN	9,585

OLA ELECTRIC remained the highest-selling 2W maker in both FY2023 and FY2024. TVS and ATHER entered the FY2024 top three, while the bottom-three composition also changed.

4-Wheeler CAGR – Top 5 by FY2024 Sales

Manufacturer	FY2022	FY2023	FY2024	CAGR
Tata Motors	12,708	28,046	48,181	94.71%
Mahindra & Mahindra	4,042	13,805	23,346	140.33%
MG Motor	1,647	3,277	8,829	131.53%
BYD India	33	920	1,466	566.52%
Hyundai Motor	110	576	1,390	255.48%

5. State Market Analysis

FY2024 EV Penetration – Top 5 States by Vehicle Category

Category	State	EV Sales	Penetration
2-Wheelers	Goa	9,768	17.99%
2-Wheelers	Kerala	64,769	13.52%
2-Wheelers	Karnataka	148,111	11.57%
2-Wheelers	Maharashtra	183,052	10.07%
2-Wheelers	Delhi	38,094	9.40%
4-Wheelers	Kerala	9,169	5.76%
4-Wheelers	Chandigarh	1,020	4.50%
4-Wheelers	Delhi	8,630	4.29%
4-Wheelers	Karnataka	12,878	4.26%
4-Wheelers	Goa	1,031	4.25%

FY2022–FY2024 Penetration Decline

State	Category	FY2022	FY2024	Change
Andaman & Nicobar Island	4-Wheelers	1.88%	0.84%	-1.04 pp
Ladakh	2-Wheelers	4.48%	4.06%	-0.41 pp

The cleaned dataset contains two state/category combinations with a negative penetration change: Andaman & Nicobar Island 4W and Ladakh 2W.

Delhi vs Karnataka – FY2024

State	Category	EV Sales	Total Vehicles	Penetration
Delhi	2-Wheelers	38,094	405,218	9.40%
Delhi	4-Wheelers	8,630	201,130	4.29%
Karnataka	2-Wheelers	148,111	1,279,767	11.57%
Karnataka	4-Wheelers	12,878	302,221	4.26%

6. State Growth & Seasonality

Top 10 States by EV Sales CAGR – FY2022 to FY2024

State	FY2022 EV Sales	FY2024 EV Sales	CAGR
Meghalaya	4	133	476.63%
Tripura	28	304	229.50%
Nagaland	1	9	200.00%
Chandigarh	411	2,877	164.58%
Chhattisgarh	4,534	28,540	150.89%
West Bengal	2,685	16,864	150.62%
Goa	1,778	10,799	146.45%
DNH and DD	35	198	137.85%
Uttar Pradesh	10,222	57,758	137.70%
Madhya Pradesh	7,916	43,223	133.67%

States with zero FY2022 EV sales do not have a finite CAGR and are excluded from this ranking. Very high CAGR values can also result from a small starting base.

EV Sales Seasonality

Across FY2022–FY2024, **March** recorded the highest aggregate monthly EV sales (291,587 units), while **June** recorded the lowest (106,709 units).

7. 2030 Projection Scenario

This is a mechanical scenario using FY2024 sales and historical FY2022–FY2024 CAGR for the 10 states with the highest FY2024 overall EV penetration. It is not a forecast; it assumes the historical CAGR continues unchanged through 2030.

State	FY2024 Pen.	CAGR	FY2024 EV	2030 Scenario
Goa	13.75%	146.45%	10,799	2,419,574
Kerala	11.59%	132.83%	73,938	11,779,401
Karnataka	10.18%	93.24%	160,989	8,383,406
Maharashtra	8.60%	101.89%	197,169	13,351,146
Delhi	7.71%	68.10%	46,724	1,054,259
Chandigarh	6.37%	164.58%	2,877	986,811
Odisha	6.33%	102.94%	39,118	2,732,814
Chhattisgarh	5.67%	150.89%	28,540	7,118,219
Tamil Nadu	5.49%	59.95%	94,314	1,579,547
Puducherry	5.37%	105.44%	3,098	232,936

8. Growth & Revenue Analysis

EV unit growth by category

Category	FY2022	FY2023	FY2024	FY22→FY24 Growth	FY23→FY24 Growth
2-Wheelers	252,573	727,903	932,692	269.28%	28.13%
4-Wheelers	18,577	47,465	86,901	367.79%	83.08%

Revenue methodology: Revenue growth requires average unit-price assumptions for 2W and 4W. The supplied project materials do not specify those prices, so this report does not invent them. With constant prices, revenue growth equals unit-sales growth.

9. Power BI Dashboard

Page 1 – India's EV Market Overview: KPI cards, EV sales trend, 2W vs 4W market mix, top states by penetration, state opportunity matrix and penetration change by state.

Page 2 – Competition & Future Opportunity: quarterly 2W sales trend for top manufacturers, top 5 4W manufacturers by year, and state growth/opportunity analysis using CAGR and penetration.

10. Key Findings

- Total EV sales rose from 271,150 units in FY2022 to 1,019,593 units in FY2024.
- Two-wheelers dominate the EV sales mix.
- OLA ELECTRIC led 2W sales in both FY2023 and FY2024.
- EV penetration differs materially across states and between 2W and 4W categories.
- State CAGR reveals fast-growing markets, but high CAGR can be driven by a small starting base.
- March was the highest-sales month and June the lowest in the combined three-year dataset.

11. Business Recommendations

- Segment the Indian market by state and vehicle category rather than treating it as one uniform market.
- Evaluate state opportunity using both penetration and growth.
- Track 2W and 4W competition separately.
- Combine sales analysis with current subsidy, charging-infrastructure and consumer research before making investment decisions.
- Use CAGR projections as scenarios, not standalone forecasts.

12. Conclusion

The project demonstrates an end-to-end Data Analyst workflow: Python data preparation, SQL analysis, DAX/Power BI modeling and business-focused visualization. It provides a structured view of EV market size, competition, state adoption, growth and future opportunity.